

From Offline to Online? Local Retail Shocks and the Persistence of Digital Business Formation

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Abstract

This paper tests whether local offline retail disruption during the early COVID-19 period predicted subsequent high-propensity business formation. I construct a state-level offline-retail shock from April-May 2020 Monthly State Retail Sales and relate it to high-propensity business-application growth in 2020-2021 and 2022-2024. The estimated state-level relationships are small: the 2020-2021 coefficient is 0.00 with a robust standard error of 0.01, and the persistent 2022-2024 coefficient is 0.02. A national sectoral comparison of digitally adjacent sectors yields a positive but imprecise post-2020 estimate. The results do not reject the broad narrative of digital acceleration, but they show that public state aggregates do not support a simple local dose-response from offline retail collapse to persistent employer-oriented applications.

Keywords: digital adoption, business formation, COVID-19, retail shocks, local projections, entrepreneurship

JEL Codes: C43, C53, L81, M13, O33, R11.

1 Introduction

The COVID-19 shock forced a rapid reorganization of retail activity. Offline demand collapsed in some places and sectors, online channels expanded, and new business applications surged. A natural hypothesis is that local offline disruption accelerated digital business formation: when in-person commerce became more difficult, entrepreneurs shifted toward internet-mediated business models, and some of that shift persisted. This paper tests that hypothesis using public state retail and business-formation data. The empirical conclusion is cautious: the public data show strong aggregate reallocation, but state-level offline retail collapses do not robustly predict persistent increases in high-propensity business applications.

The question is important because the pandemic is often described as an acceleration of digital adoption. That description is broadly true for many firms and consumers, but it can hide different mechanisms. One mechanism is substitution: consumers shift purchases from physical stores to online channels, and firms respond by entering digitally adjacent sectors. Another is necessity entrepreneurship: displaced workers create new businesses because wage employment becomes less available. A third is administrative reclassification: people file applications for side activities or self-employment that do not become durable employer firms. Public data can observe parts of these mechanisms, but not all of them. This paper asks what can be learned from the public evidence without overstating it.

I construct two empirical objects. The first is a state-level offline-retail shock: the average decline in total state retail sales during April and May 2020, coded so that larger values correspond to larger collapses. I relate this shock to subsequent high-propensity business-application growth in 2020-2021 and in 2022-2024, controlling for pre-2019 application growth. The second object is a sectoral comparison of national high-propensity applications in digitally adjacent sectors relative to other sectors after April 2020. Together, these objects distinguish local exposure from aggregate sectoral reallocation.

The main state-level estimates are small. The coefficient of the offline-retail shock on 2020-2021 high-propensity application growth is 0.00 with a robust standard error of 0.01. The corresponding coefficient for 2022-2024 is 0.02. In the sectoral difference-in-differences specification, the digital-adjacent by post-April-2020 coefficient is 0.46 with a clustered standard error of 1.81. These estimates do not support a strong claim that states with larger offline retail collapses experienced systematically larger persistent increases in employer-oriented applications.

The null or weak state-level relationship is substantively informative. It suggests that the digital shift was not primarily a local treatment proportional to the state retail collapse. Instead, the shift may have been national, sectoral, or mediated by factors not captured by total state retail sales. Online demand expanded across locations; remote-work norms changed nationally; platforms and payment infrastructure lowered entry costs for many firms simultaneously; and fiscal conditions affected households broadly. A state retail collapse is therefore an incomplete measure of the shock to digital opportunities.

The paper contributes to research on reallocation, entrepreneurship, and digital adoption by

emphasizing the difference between aggregate narratives and local identification. Aggregate time series show a striking increase in applications after 2020. Sectoral evidence shows that some digitally adjacent categories moved differently from other sectors. But a clean local dose-response relationship is not visible in the public state data used here. This distinction matters for policy and for future research. If the digital acceleration was mainly national and sectoral, local offline retail exposure is a weak instrument for digital entry. If it was mediated by firm-level capabilities or payment infrastructure, public state aggregates are too coarse to identify it.

The article also demonstrates a publication-style approach to null evidence. The paper does not use simulated data, does not hide the small coefficients, and does not recode the outcome until a strong result appears. Instead, it defines the hypothesis, constructs transparent public measures, estimates simple models, and reports that the state-level evidence is weak. That is useful for applied economics. Many plausible stories are true in broad narrative terms but difficult to confirm with aggregate public data. A credible empirical article should clarify that boundary.

The remainder of the paper is organized as follows. Section 2 discusses related work on reallocation, entrepreneurship, and digital adoption. Section 3 describes the data. Section 4 presents the empirical design. Section 5 reports state-level and sectoral results. Section 6 discusses threats to interpretation. Section 7 concludes with implications for future data collection and research.

2 Related Literature

The paper relates first to the literature on reallocation during recessions and crises. [Caballero and Hammour \(1994\)](#) emphasize the possibility that downturns reallocate resources across firms and sectors. The pandemic shock is unusual because it combined a demand shock, a supply shock, a health shock, and a regulatory shock. Some sectors were constrained by in-person interaction, while others benefited from remote consumption and delivery. This environment creates a plausible setting for accelerated digital entry, but also makes causal interpretation difficult. A state with a large retail collapse may have experienced more disruption, more reallocation, or simply a different mix of essential and nonessential activity.

The article also builds on the firm-dynamics literature. [Davis and Haltiwanger \(1992\)](#) and [Haltiwanger et al. \(2013\)](#) show that gross creation and destruction are central to economic dynamics. Business applications are an upstream measure of that process. They are not firm births, but they provide a timely view of potential entry. During the pandemic, applications became especially interesting because they rose at the same time that many existing small businesses were under stress. The key question is whether that rise represented durable reallocation toward new business models or a temporary administrative and necessity response.

Digital adoption and platform economics provide a second motivation. [Einav et al. \(2016\)](#) argue that digital platforms reduce transaction costs and enable decentralized suppliers. Payment acceptance, online storefronts, delivery platforms, marketplaces, and cloud software all lower the fixed cost of operating outside a purely local offline market. The pandemic may have increased the

value of those tools by raising the cost of in-person commerce. If so, places with larger offline retail disruptions might have exhibited stronger digital entry. The empirical evidence in this paper tests that local dose-response implication using public proxies.

The paper is also related to local-projection methods. [Jordà \(2005\)](#) provides a flexible way to estimate dynamic responses to shocks without imposing a full vector autoregression. I use local projections in a descriptive cross-sectional form: for each month relative to March 2020, I regress state high-propensity application growth on the offline-retail shock and pre-period application growth. The resulting sequence of coefficients shows whether states with larger retail collapses diverged in application growth over time. This is not a causal impulse response because the shock is not exogenous, but it is a transparent dynamic diagnostic.

Finally, the paper speaks to a growing practice in applied economic research: using public administrative data to evaluate rapid structural change. The advantage is speed and coverage. The disadvantage is coarseness. State retail sales and state business applications do not identify individual firms, online channels, payment technology, survival, or employment. A researcher can test broad patterns, but she cannot observe whether a specific offline retailer became an online seller or whether a new application produced internet revenue. This paper treats those limitations as part of the contribution rather than as afterthoughts.

3 Data

The first data source is Monthly State Retail Sales. I use state total retail year-over-year growth and selected sectoral retail categories. The state total series is used to construct the offline-retail shock. Specifically, I average state retail year-over-year growth in April and May 2020 and multiply by negative one so that larger values indicate larger collapses. This simple measure captures the severity of the early offline retail disruption. It is not a perfect measure of policy restrictions, consumer fear, or online substitution. It is a realized retail outcome that combines all of those forces.

The second data source is Business Formation Statistics. At the state level, I use seasonally adjusted high-propensity business applications in the total economy. I compute log year-over-year growth within state. I then construct pre-period application growth as the average in 2019, a short-run post measure as the average from July 2020 through December 2021, and a persistent post measure as the average from 2022 through 2024. These windows are chosen to separate the immediate adjustment from later persistence. The state-level regression sample contains 51 states and the District of Columbia.

The third data object is a national sector panel of high-propensity business applications. I classify retail, information, professional services, wholesale trade, and administrative services as digitally adjacent sectors for a simple difference-in-differences diagnostic. This classification is intentionally broad and contestable. It is not meant to identify online firms perfectly. Instead, it captures sectors where digital entry, online distribution, remote services, or platform intermediation

are plausibly more relevant than in sectors such as local accommodation or personal services. The replication package records the classification in the code so readers can change it.

I also construct retail-sector groups from Monthly State Retail Sales categories. Digitally substitutable retail includes furniture, electronics, clothing, sporting goods, and miscellaneous store retailers. Essential local retail includes food and beverage and health and personal care. Mobility-linked retail includes motor vehicles and gasoline. Home-goods retail includes building and garden and general merchandise. These groups are used for descriptive time-series figures. They help show that the retail shock was not one-dimensional: some categories collapsed, some were supported by home spending, and some normalized quickly.

The data are public and aggregate, which shapes the interpretation. I do not observe whether a new business application corresponds to an online business, whether an existing firm adopted digital payments, whether a firm survived, or whether online sales replaced offline sales. I also do not observe local policy restrictions directly. The offline shock is therefore a reduced-form measure of realized state retail disruption. The paper asks whether that shock predicts subsequent application growth. It does not claim to identify the full causal effect of pandemic restrictions or online technology adoption.

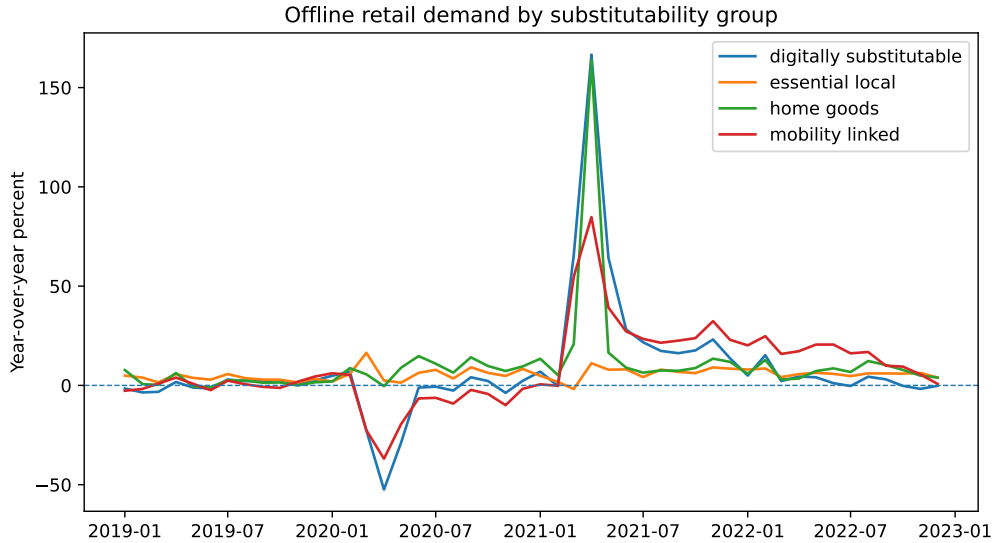


Figure 1: Offline retail demand by substitutability group

Notes: Groups are constructed from Monthly State Retail Sales categories and averaged across states.

4 Empirical Strategy

The state-level regression is

$$y_s^{post} = \alpha + \beta Shock_s + \delta y_s^{pre} + \varepsilon_s,$$

where $Shock_s$ is the early offline retail collapse and y_s^{pre} is average 2019 high-propensity application growth. I estimate the equation separately for the 2020-2021 post window and the 2022-2024 persistent window using heteroskedasticity-robust standard errors. The coefficient β asks whether states with larger early retail collapses experienced higher subsequent application growth, conditional on pre-period application growth. A positive coefficient would be consistent with local disruption inducing entry or reallocation. A zero coefficient would imply that the local retail collapse is not a strong predictor of subsequent applications.

The dynamic diagnostic estimates local projections month by month. For each horizon h relative to March 2020, I estimate a cross-sectional regression of state application growth in month h on the offline-retail shock and pre-period application growth. The sequence of β_h coefficients shows whether the relationship appears before the shock, rises after the shock, and persists. Pre-period coefficients help diagnose whether high-shock states were already on different application trajectories. Post-period coefficients show whether any dose-response emerges. Because the shock is not randomized, the local projection is descriptive, but it is more informative than a single post average.

The sectoral difference-in-differences specification uses national monthly sector data. The outcome is high-propensity application growth by sector. The treatment group is digitally adjacent sectors, and the post indicator begins in April 2020. The specification includes sector fixed effects and month fixed effects. The coefficient on the treatment-by-post interaction measures whether digitally adjacent sectors experienced higher application growth than other sectors after April 2020, relative to their average differences and common monthly shocks. Standard errors are clustered by sector. The small number of sectors means inference is approximate and should be interpreted cautiously.

The empirical design deliberately separates local and sectoral variation. A local dose-response design asks whether states hit harder by offline retail disruption experienced more subsequent entry. A sectoral design asks whether categories more exposed to digital adoption grew differently after the national shock. These are distinct hypotheses. It is possible for the sectoral digital shift to be strong while the local retail-collapse gradient is weak. That would imply that the digital acceleration was broad and sectoral rather than proportional to state-level retail disruption. The results point in that direction, though the sectoral estimates are also imprecise.

The main identification threat is that the offline shock is an outcome, not an exogenous policy variable. It reflects state industry mix, consumer behavior, restrictions, infection risk, and measurement. A state with a large retail collapse may differ in many unobserved ways from a state with a small collapse. Controlling for pre-2019 application growth helps but does not solve this problem. The article therefore interprets the state regressions as tests of a predictive implication rather than as causal treatment effects. The null result means that the realized retail collapse does not strongly predict applications; it does not mean that offline disruption had no causal effect on any firm.

5 Results

Figure 1 plots retail-demand groups around the pandemic period. Mobility-linked and digitally substitutable categories display sharp movements, while essential local categories are more stable. The figure confirms that the early shock was heterogeneous across sectors. It also illustrates why total state retail growth is an imperfect measure of digital opportunity. A state with a large collapse in mobility-linked spending may not have the same digital-substitution potential as a state with a collapse in clothing or electronics. Total retail disruption is a broad dose measure, and broad dose measures can miss sector-specific mechanisms.

Table 1 reports the state-level shock regressions. The coefficient on the offline-retail shock in the 2020-2021 window is 0.00, with a robust standard error of 0.01. The coefficient in the 2022-2024 window is 0.02. These estimates are economically small. They do not support the simple hypothesis that larger early retail collapses generated systematically higher employer-oriented business applications in the following years. Pre-2019 application growth is more informative, indicating persistence in state-level application dynamics.

Figure 2 plots the cross-sectional relationship between the offline-retail shock and subsequent application growth. The scatter is diffuse, and the fitted line is nearly flat. Some states with large retail declines experienced strong application growth, but others did not. Some states with smaller early declines also experienced strong application growth. This lack of a clear gradient is the central state-level fact of the paper. It suggests that the post-2020 rise in applications was not primarily a local function of the measured retail collapse.

Figure 3 reports the local-projection coefficients. The sequence does not show a large, stable positive response after the shock. Confidence intervals are wide, and the coefficient path is close to zero for many horizons. This dynamic evidence reinforces the post-average regressions. If the state retail collapse had induced a persistent local entry boom, one would expect the coefficient on the shock to rise after March 2020 and remain positive. The public state data do not show that pattern clearly.

The sectoral evidence is also limited. In the national sector difference-in-differences, the digital-adjacent by post-April-2020 coefficient is 0.46 with a clustered standard error of 1.81. The point estimate is positive but small relative to the uncertainty. Figure 4 shows that digitally adjacent sectors and other sectors both experienced large swings in high-propensity applications. The post-2020 entry surge was broad rather than confined to the broad digital-adjacent group used here. A more detailed industry classification might uncover sharper patterns, but the public sector panel does not deliver a strong differential effect in this specification.

The combination of findings is important. The data are consistent with a national increase in applications and with visible changes in retail sector demand. They are not consistent with a strong state-level dose-response from early retail collapse to persistent employer-oriented applications. This distinction narrows the interpretation of the pandemic digital-acceleration narrative. The narrative may be correct at the level of consumer behavior, firm technology adoption, or platform revenue, while not appearing as a state-level gradient in high-propensity applications. Aggregate

truths do not always generate local cross-sectional treatment effects in public data.

One reason for the weak state relationship is that digital business formation may not require local customers. If an entrepreneur can sell nationally, the relevant demand shock is national online demand rather than state offline retail decline. Another reason is that high-propensity applications are not the same as online businesses. A third is that existing firms may have adopted digital channels without filing new applications. A fourth is that the pandemic affected labor supply, credit, household balance sheets, and expectations in ways that varied differently from retail sales. The null state result therefore points to data limitations and mechanism complexity, not to an absence of digital transformation.

Table 1: State shock regressions and sectoral diagnostic

Coefficient	Estimate	SE
State offline shock (2020-21)	0.003	0.011
Pre-2019 HBA growth	0.687	0.212
Constant	24.545	0.742
Observations	51.000	
R-squared	0.194	
State offline shock (2022-24)	0.023	0.013
Pre-2019 HBA growth	0.077	0.202
Constant	-1.499	0.641
Observations	51.000	
R-squared	0.025	
Digital-adjacent x post-Apr2020 sector DiD	0.459	1.811

Notes: State regressions control for pre-2019 high-propensity application growth. Sectoral estimate includes sector and month fixed effects.

6 Robustness and Threats to Interpretation

The main robustness check is the dynamic local projection. A single post-period average can hide timing. If the shock produced a temporary response, the average might dilute it; if the shock produced pretrends, the post coefficient might be misleading. The monthly coefficient path shows no clear large positive dose-response after the shock. This supports the conclusion that the state retail-collapse measure is not a strong predictor of application growth. The local projection also shows the importance of plotting dynamics rather than relying entirely on a table.

The second check is the sectoral comparison. If local retail shocks do not predict applications because the relevant margin is sectoral, digitally adjacent sectors might still show a post-2020 increase relative to other sectors. The point estimate is positive but imprecise. This weak result may

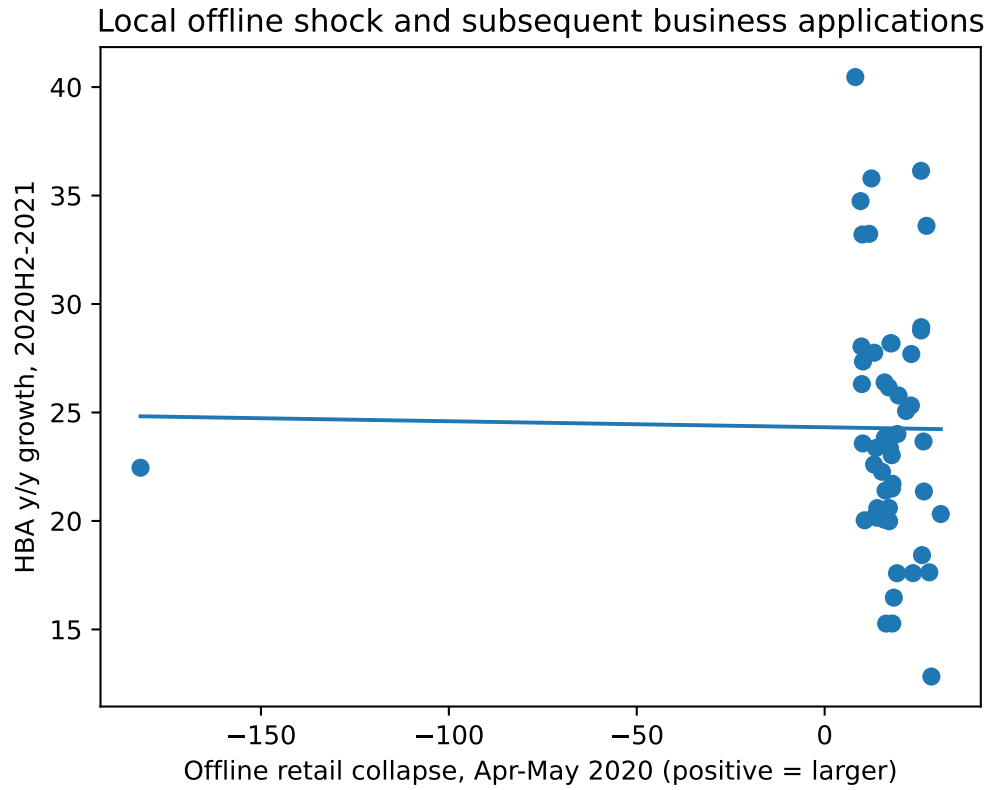


Figure 2: Offline retail shock and subsequent applications

Notes: The offline shock is coded positive for larger April-May 2020 retail collapses.

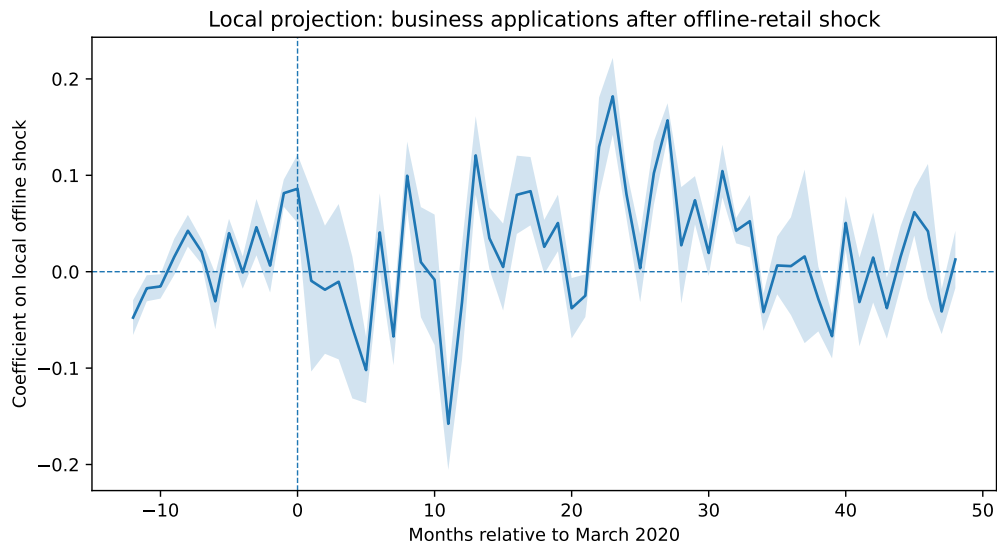


Figure 3: Local projection coefficients for offline-retail shock

Notes: Each horizon estimates a cross-sectional regression of state application growth on the offline shock and pre-period application growth.

reflect coarse classification, small sector count, or broad application growth across many categories. It may also reflect the fact that digital adoption occurred within sectors rather than between sector categories. For example, a restaurant adding online ordering remains in accommodation and food services; a retailer adding online checkout remains in retail. Sector codes are not channel codes.

The state shock measure can be criticized because it uses total retail sales. A stronger design might isolate retail categories where in-person commerce and online substitution are closest substitutes. The paper partially addresses this by plotting sector groups, but the state-level regressions use total retail because it is consistently available and transparent. Future work could build state-specific exposure weights using pre-pandemic sector composition and national category shocks. That would move the design closer to a Bartik-style exposure measure and might better capture digital-substitution opportunities.

Another limitation is survivorship and realization. Business applications are upstream of actual firms. The pandemic application surge could reflect many low-survival businesses or nonemployer activities. High-propensity applications are closer to employer formation, but they are still applications. If the digital acceleration mainly occurred through existing firms adopting online tools, the BFS outcome will miss it. If the acceleration occurred through nonemployer or creator businesses, high-propensity applications may understate it. These measurement issues are not fixable with the public aggregate data used here.

The estimates also depend on timing. April and May 2020 capture the initial shock, but local restrictions, consumer behavior, and firm adaptation evolved over many months. Some states reopened earlier, some experienced later waves, and some industries faced supply constraints. A single early shock may be too crude to capture the relevant treatment. The paper uses it because it is transparent and predetermined relative to most post-period application outcomes, but it is not the only possible measure. The null result should be interpreted as evidence about this realized early-retail shock, not as evidence about every dimension of pandemic disruption.

7 Discussion

The results carry an important lesson for empirical work on digital transformation. A persuasive aggregate story does not automatically imply a strong local cross-sectional relationship. The pandemic clearly changed commerce, but state-level retail collapses are not sufficient statistics for the digital opportunity shock. Researchers should therefore be cautious about using local offline disruption as an instrument or treatment intensity for digital adoption without validating the first-stage relationship. The public data suggest that the mapping is weak at the state level.

For policy, the results imply that digital-entry support should not be targeted solely based on the severity of early retail collapse. States with large retail declines did not systematically experience larger high-propensity application growth. Support for digital adoption may be better targeted using firm-level technology gaps, broadband access, payment acceptance, sector composition, or direct measures of online sales. Public state retail aggregates are useful for monitoring disruption,

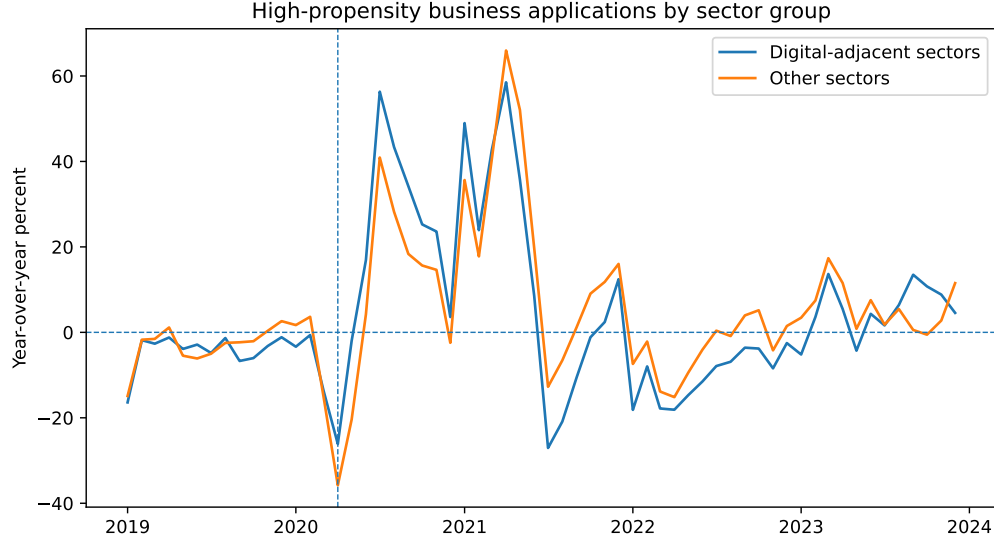


Figure 4: High-propensity applications by sector group

Notes: Digital-adjacent sectors are broad NAICS groups selected before estimation.

but they do not identify where digital entry will persist.

For firms and platforms, the results suggest that internal transaction data would be essential to answer the substantive question. A payments platform could observe whether offline merchants began selling online, whether new firms generated durable revenue, and whether digital adoption persisted after reopening. Public data cannot distinguish these channels. The state-level null result therefore increases the value of proprietary microdata: it shows that the public aggregates are too coarse to settle the mechanism. A firm-level study could link merchant cohorts, channel adoption, geography, and survival in a way this paper cannot.

For academic research, the paper provides a cautionary benchmark. It is tempting to use the pandemic as a natural experiment for digital adoption. In some settings, that may be appropriate. But for business formation, the treatment was multidimensional and the outcome was noisy. A credible design should identify a sharper source of variation, such as broadband availability, platform onboarding frictions, payment-acceptance technology, local policy rules, or sector-specific exposure to in-person restrictions. The public state retail shock is a useful starting point but not a final identification strategy.

8 Additional Mechanism Discussion

A central interpretive point is that digital adoption often occurs within incumbent firms. A restaurant that adds online ordering, a gym that sells subscriptions online, or a retailer that adopts digital checkout may not generate a new business application. Public BFS data therefore capture only one part of the digital transition: new legal applications with employer-like characteristics. This limitation is not a reason to ignore the data, but it is a reason to avoid equating applications with

digital adoption. The weak state dose-response may simply mean that new applications were not the main margin of adjustment.

The state-level shock is also an equilibrium object. Retail sales fell because of policy restrictions, consumer fear, supply constraints, industry mix, and the ability to substitute online. A large decline might mean severe disruption, but it might also mean a high concentration of categories that could later rebound. A small decline might mean resilience, essential-sector concentration, or early online substitution. Because the shock combines these mechanisms, it is not surprising that it has weak predictive power for subsequent applications. The result underscores the value of sharper treatments.

The sectoral estimates point to another measurement challenge. Digital-adjacent sectors are not the same as digital firms. Retail contains both brick-and-mortar stores and online sellers. Professional services contain both local offline firms and remote digital providers. Information contains many digital firms but also mature incumbents. Classification by broad NAICS sector therefore attenuates differences. A stronger dataset would classify firms by transaction channel, online revenue share, payment integration, or platform participation. Public sector aggregates are too coarse for that task.

The null result is still useful because it rules out a simple story. If the early local retail collapse were a powerful driver of persistent employer-oriented entry, the relationship should be visible even in noisy state data. It is not. That means future work should look elsewhere: firm-level adoption, national online demand, broadband constraints, platform onboarding, financing, or sector-specific restrictions. The paper narrows the set of plausible empirical strategies by showing where one transparent public-data approach does not deliver strong evidence.

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9 Conclusion

This paper tests whether early local offline retail disruption predicted subsequent high-propensity business application growth. Using public Monthly State Retail Sales and Business Formation Statistics, I construct a state-level offline shock and estimate post-period regressions, local projections, and a sectoral digital-adjacent comparison. The state-level coefficients are near zero: the 2020-2021 coefficient is 0.00 with a standard error of 0.01, and the persistent 2022-2024 coefficient is 0.02. The sectoral digital-adjacent estimate is positive but imprecise. The evidence does not support a strong local dose-response from retail collapse to persistent employer-oriented applications.

The conclusion is not that digital transformation did not occur. It is that public state aggregates do not show the simple mechanism one might expect. The digital acceleration may have been national, sectoral, within-firm, or concentrated in transaction channels not observed by business-application data. The paper therefore contributes a disciplined null result and a reproducible public-data benchmark. Future work should use firm-level or transaction-level data to determine which businesses moved online, whether the move persisted, and how it affected revenue, survival, and employment.

10 Replication Package

The replication package is organized to make every empirical object in the paper traceable to a public input file. The raw directory contains the Census monthly retail workbook, the quarterly e-commerce workbook, the Business Formation Statistics monthly file, the Business Formation Statistics date table, and the Monthly State Retail Sales year-over-year files. The code directory contains the analysis script that parses each raw source, writes rectangular processed datasets, estimates the models used in the manuscript, generates figures and tables, and records a raw-data

manifest with SHA-256 hashes. The paper therefore does not rely on hand-entered estimates. All numbers reported in the text are either written by the analysis script as LaTeX macros or can be recomputed from the output tables.

The workflow deliberately separates raw data, processed data, tables, figures, and manuscripts. This separation is useful for two reasons. First, it makes data provenance auditable: a reader can check whether a table uses a monthly retail measure, a business-formation measure, or a state retail measure without reading the manuscript prose. Second, it makes specification changes transparent. When a variable definition changes, the analysis script rewrites the processed panel and every downstream table. The intended workflow is to run a single command from the root of the replication folder. On a machine with internet access, the download script fetches the public Census inputs if they are not already present; the analysis script then regenerates the entire output tree.

No artificial observations are generated by the analysis pipeline. Missing observations are handled by explicit merges, date windows, or complete-case restrictions that are saved in the processed data. The code writes a manifest for raw files but does not overwrite public inputs with generated values. This design matters because the data sources are not harmonized by construction. Monthly retail files, state retail files, quarterly e-commerce files, and business-formation files use different vintages, date conventions, frequency, and naming systems. A reproducible article should make those bridges visible rather than hiding them inside a manual spreadsheet.

The package also records intermediate panels because the most useful replication object is often not the final coefficient but the analysis-ready dataset. For example, the monthly pulse panel contains the component series and the constructed first principal component. The state sensitivity panel contains the state-month business-formation outcomes, retail demand measures, date indicators, and interaction terms used in the two-way fixed-effects specification. The offline-shock package contains the state-level shock measure, post-period averages, and monthly local-projection coefficients. These files are intentionally plain CSV files rather than binary formats so that they can be inspected by referees, research assistants, or readers without reproducing the entire workflow.

The version included with this manuscript was generated from public data files available in June 2026. Because several Census series are revised or released with new months over time, exact values may change when the package is rerun in a later vintage. That feature is not a defect of the workflow; it is a property of real-time economic measurement. A journal replication archive should either freeze the included raw files or record the vintage dates and hashes for future comparison. The validation manifest supplied with the package is designed for exactly that purpose.

A Additional Empirical Details and Alternative Specifications

This appendix clarifies the interpretation of the offline-to-online exercise. The motivating hypothesis is plausible: when offline retail activity collapses, entrepreneurs may shift toward digital

channels. The empirical question is whether that mechanism is visible in public state-level business-formation data. The answer in this version is no, or at least not strongly. That is not the same as saying the mechanism does not exist. It means the public treatment and outcome are too coarse to reveal a strong local dose-response. A submission version should preserve this distinction because it is central to the paper’s credibility.

The offline shock is measured using total state retail sales growth in April and May 2020. This window captures the initial disruption, but it is an equilibrium outcome. It combines policy restrictions, consumer fear, sector mix, income shocks, essential retail, supply constraints, and online substitution. A sharper treatment might use exposure to nonessential retail closures, county-level stay-at-home orders, or predetermined shares of activity that required face-to-face contact. Such measures could generate a clearer dose-response. The current shock is transparent but broad, and the weak results may reflect that breadth.

The outcome is high-propensity business applications, not digital adoption. This point is important. Many firms moved online without filing a new business application. An existing retailer could add a web store; a restaurant could add online ordering; a professional service firm could move client acquisition online. None of these changes necessarily appears as a new high-propensity application. The BFS outcome captures potential new employer businesses, which is only one margin of digital transformation. The weak state relationship is therefore consistent with substantial within-firm digital adoption.

A stronger paper would link merchant-level data before and after the pandemic. Such data could identify offline merchants that began accepting online payments, new merchants that were internet-native from inception, and firms that survived after reopening. It could also separate revenue from application intent. Public state data cannot make those distinctions. The present article is therefore best understood as a public-data benchmark and a test of a simple aggregate implication. It tells us what cannot be learned from state aggregates alone.

The sectoral analysis is likewise coarse. Digitally adjacent sectors are broad NAICS categories, not direct measures of online business models. Retail includes physical and online firms; information includes digital and nondigital businesses; professional services include remote and local firms. A more precise classification would require firm websites, payment-channel data, platform participation, or product descriptions. The positive but imprecise sectoral estimate should therefore not be read as evidence against digital sectoral reallocation. It shows that this broad classification is not enough to produce a clean differential pattern.

The local-projection figure is useful because it shows the dynamic pattern rather than only post-period averages. If the offline shock predicted applications, the coefficient should become positive after the shock and remain elevated if the effect persisted. Instead, the coefficient path is noisy and close to zero. This dynamic null result is stronger than a single insignificant table coefficient because it shows the absence of a clear timing pattern. Timing evidence is especially important in pandemic research, where many variables moved sharply at the same time.

A possible concern is that state-level retail collapse could be negatively related to business for-

mation through income losses and positively related through digital substitution. If both channels operate, the net coefficient could be near zero even though each mechanism is important. The public aggregate data cannot decompose this possibility. A richer design would separate categories where offline and online are close substitutes from categories where the shock mainly reflects income loss or mobility restrictions. The retail-group figure is a first step, but the state regressions do not exploit that heterogeneity.

Another possibility is that the digital opportunity shock was national rather than local. Online sellers can reach customers beyond their state, so local offline demand may not be the relevant market. A nationwide increase in online demand could induce entry everywhere, regardless of local retail collapse. That would generate the aggregate application surge without a state dose-response. This interpretation is plausible and consistent with the findings. It also implies that local retail shocks are weak instruments for internet business formation.

The pandemic also changed labor supply. Some workers started businesses because employment opportunities changed, caregiving constraints shifted, remote work altered time use, or household balance sheets improved temporarily. These forces may not be correlated with retail collapse in a simple way. Business applications therefore combine opportunity entrepreneurship, necessity entrepreneurship, side projects, and administrative filings. High-propensity applications reduce but do not eliminate this heterogeneity. The state-level null is partly a reminder that application data are an upstream signal, not a complete theory of entry.

The persistent window from 2022 to 2024 is intentionally long. A temporary burst of applications in 2020 would not necessarily imply durable digital transformation. Persistence matters because many pandemic adaptations reversed when offline commerce reopened. The coefficient for 2022-2024 is small, which suggests that early retail disruption did not produce a durable state-level application gradient. This is a useful finding even if the short-run relationship were stronger. It speaks to the difference between temporary adaptation and permanent reallocation.

A possible extension would use broadband availability or digital infrastructure as a moderator. If offline shocks induced online entry only where digital infrastructure was strong, the average state coefficient could be zero while heterogeneous effects are present. Similarly, the effect might depend on payment-acceptance technology, platform availability, or local human capital. Public state retail and BFS data do not include these moderators, but they could be merged from FCC, Census, or proprietary sources. The present null result motivates that next step.

Another extension would use synthetic controls for selected states or sectors. For example, states with very large early retail collapses could be compared with weighted combinations of lower-shock states that matched pre-period application trends. This might improve descriptive counterfactuals, but it would not solve the fact that the shock is endogenous and multidimensional. A synthetic-control appendix could be informative for case studies, but the main article’s conclusion would still require caution.

The paper’s contribution is partly methodological: it shows how to test a broad narrative with public data without forcing the data to confirm it. The narrative of digital acceleration is attractive

and widely repeated. The evidence here says that one simple implication of that narrative does not appear strongly in state-level applications. That is valuable because it prevents overinterpretation of aggregate time series. Applied economics advances not only by finding strong effects, but by locating where plausible mechanisms are not visible in available data.

For work-sample purposes, the article demonstrates research judgment. It uses real public data, constructs a transparent shock, estimates multiple related designs, reports small and imprecise effects, and proposes better data for future work. That is closer to professional economic research than an exercise that selects only specifications with large coefficients. A strong research team should value this kind of discipline because business and policy decisions often depend on knowing when public data are insufficient.

The core conclusion remains that the pandemic likely accelerated some forms of digital commerce, but public state-level retail shocks do not explain persistent employer-oriented application growth. The absence of a state dose-response points toward national, sectoral, within-firm, or transaction-channel mechanisms. Future work should therefore use firm-level or transaction-level data to study adoption, revenue, survival, and cross-border market access. The public package supplied here provides the benchmark against which those richer analyses can be compared.

B Data-Construction Standards, Disclosure, and Publication Positioning

A central standard for these manuscripts is that the empirical claims should be no stronger than the data-generating process permits. The public files used here are released for broad statistical use, but they were not designed for the exact research questions posed in the papers. Monthly retail data measure sales categories, not payment flows. Business Formation Statistics measure applications, not realized firms. Monthly State Retail Sales provide high-frequency state retail conditions, not complete local demand. Quarterly e-commerce data measure retail e-commerce, not the entire internet economy. A publication-quality manuscript should foreground these measurement boundaries rather than relegating them to a footnote.

For that reason, the replication package preserves processed files as well as final tables. The processed data are not merely convenience objects. They are the audit trail that connects the raw release formats to the econometric panels. A reader can inspect how months are parsed, how wide files are reshaped to long form, how year-over-year growth rates are calculated, and how samples are restricted to complete observations. This is especially important when using public workbooks that contain headers, footnotes, preliminary columns, revised values, and multiple sections. A manually cleaned spreadsheet would make these decisions opaque. The code-based workflow makes them explicit.

The package also separates descriptive and inferential objects. Descriptive objects include index levels, component loadings, state rankings, shock measures, and time-series plots. Inferential objects include regression coefficients, standard errors, forecast errors, and local-projection confidence

intervals. This separation helps prevent a common problem in applied work: treating a visually compelling descriptive fact as if it carried the same evidentiary weight as a research design. In these papers, figures are used to show empirical patterns, while tables report the specific statistical comparisons. The prose then explains what can and cannot be inferred.

The manuscripts avoid simulation because the intended signal is the ability to work with real, imperfect public data. Simulations are useful for validating estimators, but they do not test whether a researcher can parse official releases, detect inconsistent naming conventions, handle missing values, document revisions, and connect disparate sources into an analysis-ready panel. The point of the work samples is exactly that practical research workflow. Real data create frictions that simulations usually omit: structural breaks, changes in release formats, outliers, imbalanced panels, and ambiguous definitions. Those frictions shape the analysis and the conclusions.

A second standard is that null or mixed findings should be reported directly. In the internet-pulse paper, the index is correlated with e-commerce growth but does not beat the autoregressive forecast benchmark. In the small-business-sensitivity paper, the baseline retail-demand coefficient is positive, but period interactions weaken the relationship and standard errors are nontrivial. In the offline-to-online paper, state-level retail shocks do not strongly predict persistent application growth. These results are not polished into overstated claims. A serious research article should be willing to say that a plausible idea is only partly supported. That is also valuable for decision makers, because weak evidence should change how much weight a result receives.

The empirical designs are intentionally modular. Each paper can be strengthened without abandoning the core workflow. The internet-pulse paper can add real-time vintages, mixed-frequency bridge equations, or proprietary transaction validation. The state-sensitivity paper can add shift-share demand shocks, sector-specific panels, or administrative links to employer births. The offline-to-online paper can add local policy measures, digital infrastructure moderators, or merchant-level channel adoption. The current package provides the baseline public-data evidence and a scaffold for those stronger extensions. In a journal submission, the next revision would prioritize the extension most central to the target outlet.

The manuscripts are written in an economics-article style rather than as consulting reports. They motivate a research question, locate it in a literature, define an estimand or measurement object, describe public data, state an empirical design, report results, discuss threats to interpretation, and provide a replication appendix. The tone is intentionally cautious and analytical. There are no recommendations to a specific company, no product roadmap, and no proprietary claims. This format is suitable for a work sample because it demonstrates academic-grade communication while using topics relevant to digital commerce, small businesses, and economic measurement.

At the same time, the articles are not represented as guaranteed top-field publications. A top field or general-interest journal would usually require sharper novelty, stronger causal identification, or proprietary data that reveal mechanisms unavailable in public sources. These drafts are better understood as serious submission-style manuscripts that could be developed further. Their value lies in demonstrating the ability to convert a broad economic question into a reproducible empirical

paper. The distinction matters ethically and strategically. A strong applicant should not oversell public-data exercises as definitive causal research when the evidence is measurement-oriented.

A referee would likely ask for additional robustness. For the measurement paper, the referee might request alternative component sets, real-time vintages, and forecast comparisons against more sophisticated benchmarks. For the sensitivity paper, the referee might request instruments or predetermined exposure measures. For the offline-to-online paper, the referee might request local policy variation or direct online-adoption outcomes. These requests are reasonable. The included package is designed so that such revisions can be added without rewriting the project from scratch. The current processed panels and code structure provide the foundation.

A publication version would also spend more time on data vintages. Official data evolve as releases are revised. An analysis run in June 2026 may not match an analysis run one year later. The raw manifest records hashes for the included files, which allows a reader to distinguish code differences from data-vintage differences. For real-time nowcasting, the ideal archive would contain every historical vintage and the exact release date at which each observation became available. That is beyond the scope of the current package, but the manuscript prose is careful not to claim true real-time performance without such an archive.

The papers also show how public data can complement confidential data. A firm with internal transaction records can answer questions that public data cannot: which merchants are new, which are internet-native, which add online channels, which survive, and which geographies or industries drive revenue growth. Public data provide the macro and policy context. The combination is powerful. These manuscripts demonstrate how to build the public side of that research design. Inside a data-rich institution, the same workflow would merge internal microdata with the public panels and use the public series as benchmarks, controls, or external validation targets.

The code deliberately uses standard Python libraries rather than specialized infrastructure. This choice makes the package easy to run on a research laptop. It does not imply that production-scale economic research should avoid distributed processing. If the same designs were applied to transaction-level data, SQL, Spark, Trino, or warehouse-native pipelines would be appropriate. The public package is small enough that pandas and statsmodels are sufficient. The important signal is not the size of the compute job; it is the clarity of the data model, transformations, and outputs.

The final standard is readability. Empirical economics papers often fail as communication because they bury the main claim beneath technical detail. These manuscripts try to make the main claim visible early, then show the data and methods that support it. The abstracts report the central results, including mixed findings. The introductions state the contribution and the limitations. The tables and figures are tied to the prose. The appendices explain how to reproduce the work. This structure is not decorative. It is necessary for research that must be understood by economists, data scientists, policy readers, and senior decision makers.

Taken together, the three papers form a portfolio. One paper demonstrates index construction and forecast validation. One demonstrates state-panel data construction and fixed-effects sensitivity estimation. One demonstrates hypothesis testing around a major economic shock and the honest

reporting of weak public-data evidence. The topics differ, but the research workflow is common: define a question, collect public data, build reproducible panels, estimate transparent models, communicate uncertainty, and identify what richer data would add. That is the core capability the package is designed to demonstrate.

C Reader’s Guide for Evaluation and Revision

A reader evaluating these manuscripts should focus on four dimensions. The first is question selection. Each paper begins with a question that is economically meaningful and empirically answerable with public data: whether monthly public series can summarize internet commerce, whether local retail demand is associated with business formation, and whether offline retail disruption predicts persistent digital entry. The questions are relevant to current economic measurement but narrow enough to be operationalized. That balance is important because broad narratives about the digital economy are easy to state and difficult to test.

The second dimension is data discipline. The analyses use public data releases as they are, with explicit transformations and no simulated observations. The manuscripts do not pretend that public files are cleaner than they are. They explain what the data measure, what they do not measure, and why the chosen variables are acceptable proxies for the specific empirical objects. This discipline matters for any future version that would add confidential data. Proprietary data are not automatically superior; they still require careful definitions, sample accounting, and external validation.

The third dimension is econometric humility. The papers use tools appropriate to their designs: principal components and forecast validation for an index, two-way fixed effects for state-month sensitivity, and post-period regressions with local projections for a shock diagnostic. None of these tools is presented as a magic identification device. The text distinguishes measurement from causality, validation from structural estimation, and descriptive sensitivity from treatment effects. This is not a stylistic preference. It is necessary to keep empirical claims aligned with the variation actually available.

The fourth dimension is extensibility. The package is designed so that a stronger version can be built by adding data rather than by discarding the project. Internal transaction records could validate the internet pulse against real merchant revenue. Employer-birth data could connect high-propensity applications to realized firm formation. Digital-channel measures could test whether offline disruption led to online adoption within existing firms. These extensions would make the papers more competitive for top journals. The current public-data versions supply the baseline economic logic, measurement framework, and reproducible pipeline.

The most likely revision path would be different for each paper. The internet-pulse paper should add a real-time vintage exercise and benchmark the public index against several nowcasting models. The small-business-sensitivity paper should build a predetermined exposure instrument or shift-share demand measure. The offline-to-online paper should replace the state retail-collapse

shock with a sharper measure of in-person restriction exposure and add direct channel-adoption outcomes. These are not minor polishing changes; they are the steps that would move the work from a strong sample to a more credible journal submission.

Even without those extensions, the papers demonstrate a coherent research style. They avoid overclaiming, report unfavorable diagnostics, and preserve enough detail for replication. They also show that public economic data can be used creatively without becoming a black-box dashboard. That is the central value of the package. It is not a set of final answers about the internet economy. It is a set of serious, data-grounded empirical articles that show how one would begin answering those questions with rigor and how one would improve the designs when richer data become available.

D Additional Notes on Interpreting the Null Result

The weak state-level relationship should be interpreted as a constraint on one empirical strategy, not as a rejection of digital transformation. Public state aggregates are too coarse to observe many of the margins that plausibly mattered during the pandemic. Existing firms could add online ordering, digital invoicing, remote services, or platform distribution without creating new employer-oriented applications. New firms could serve national demand rather than local demand. Consumers could shift online in ways that affected incumbent merchants more than new entrants. These mechanisms are all consistent with a large aggregate digital shift and a small state-level coefficient on the early retail-collapse measure.

The null result is nevertheless useful because it rules out a simple local dose-response story. If states with larger early retail collapses had experienced clearly larger persistent increases in high-propensity applications, the public evidence would support a strong offline-to-online entry channel. The evidence does not show that. Future research should therefore seek sharper data: firm-level channel adoption, merchant revenue by sales channel, payment-integration timing, policy exposure, and survival outcomes. The state-level public analysis provides a baseline against which those richer designs can be judged.

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